

# Maxon Internship

Reflection

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# 1. Introduction

The following document is a reflection on the internship carried out at Maxon Motor in Sachseln, Switzerland, as part of the Bachelor Applied Computer Science programme at Thomas More University of Applied Sciences. It looks back on the project from two perspectives. Chapter 2 provides a substantive reflection, evaluating what was accomplished, what value it delivers, and what remains for the future. Chapter 3 offers a personal reflection on what the internship meant on a human level: what was learned, how it challenged me, and how it contributed to my growth as a professional.

## 2. Substantive reflection

During the internship, the primary goal defined in the project plan was successfully achieved: the MIND AI pipeline is now fully operational as a ctrlX application, installable directly on the ctrlX CORE controller through the standard snap packaging format. Beyond the original scope, an additional feature was developed in collaboration with the internship mentor: a live motor interface that allows users to connect their own motor, run training and prediction cycles directly from the application, and build a reliable predictive model without relying on externally provided datasets.

For Maxon, this result means the MIND system is significantly closer to being distributable to clients. The application can be installed on any compatible ctrlX controller without additional infrastructure, and the live capture feature makes it self-sufficient in a way the original pipeline was not. For end users, the interface is accessible through the standard ctrlX web environment and supports both English and German, lowering the barrier to adoption.

The core deliverables are complete. The most notable remaining limitation is that automatically generating bad cycles by modifying the P-gain of the motor is not yet fully functional when running the application directly from the controller; this was achievable in a development setup but requires further work to be reliable in the packaged environment. The broader MIND system is still under active development, so continued version parity with the main branch will also be needed before the application is ready for a full production release.

For the future, the most natural next step is submitting the application to the ctrlX App Store, which would make it available to a wider audience beyond the internal Maxon development team. It would also be worthwhile to invest in a two-motor test setup for generating bad cycles, as this would make the physical testing methodology more controlled and reproducible than the current workaround.

### 3. Personal reflection

This internship was a deliberate choice to step outside of my comfort zone. Moving to a new country, joining a professional team working on a complex technical product, and taking full ownership of a project from the start were all experiences I actively sought out, and they delivered exactly the kind of growth I was hoping for.

On a technical level, the internship pushed me into areas I had little prior experience with: embedded systems, snap packaging, sandboxed deployment environments, and machine learning pipelines. None of these were topics I could have learned purely in a classroom setting; the understanding came from working through real constraints and real failures over several weeks. I also developed a much stronger sense of how to approach a problem systematically: when something did not work in the packaged environment but worked locally, I learned to isolate variables, read logs carefully, and ask the right questions rather than trying random fixes.

Beyond technical skills, this internship also challenged me in ways I did not fully anticipate. Working in a professional environment where decisions I made had a real impact on a product being developed for actual clients required a level of responsibility and confidence that is difficult to simulate in academic projects. I learned to communicate more effectively with colleagues from different backgrounds and to ask for help at the right moment rather than spending too long stuck on something alone.

One area I grew in more than expected was documentation. It is something I had previously underestimated, but seeing firsthand how important it is for others to be able to understand and continue your work gave me a much more genuine appreciation for writing things down clearly and consistently.

I am also grateful for the environment at Maxon itself. From the first day I was treated as a full member of the team rather than as a temporary visitor, and the willingness of my colleagues to share their knowledge and time made a significant difference in what I was able to achieve. Although I was initially uncertain about navigating a multilingual, international work environment, the warmth from the people around me made it easy to feel at home from the very first week.

Looking back, this internship confirmed that I enjoy working on technically challenging, open-ended problems in a collaborative professional setting. It gave me a clearer picture of the kind of work I want to pursue after graduating, and I leave it with both a finished product I am proud of and a set of skills I will carry forward.